

Open Science Pool (OSPool)

Free high-throughput computing via opportunistic scavenger pool — great for embarrassingly-parallel ML workloads

J.C. Vaught · April 2026

Program Overview

The Open Science Grid (OSG) operates the *Open Science Pool* (OSPool), a nationally distributed pool of opportunistic compute resources contributed by United States universities, national laboratories, and a handful of international partners. Resources are aggregated and scheduled through HTCondor, and jobs are matched to whichever site happens to have idle capacity when the user submits. The pool is explicitly *high-throughput* rather than *high-performance*. It is engineered around large numbers of small, independent jobs that can be re-run if preempted, not around tightly coupled MPI codes that require a low-latency fabric.

OSPool is free to use for United States open-science research, with no allocation request, no peer review, and no cost share. The service is governed by the OSG Partnership, a consortium led by the University of Wisconsin-Madison Center for High Throughput Computing (CHTC), the University of Chicago, the University of Nebraska-Lincoln Holland Computing Center, the University of California San Diego (UCSD), and Syracuse University, with additional contributing sites at Florida, Clemson, Purdue, Caltech, and many others. Users access the pool through an *Access Point* (AP) such as OSG Connect (`ap40.uw.osg-htc.org`), the CHTC submit nodes, or a campus AP federated into the pool.

Hardware Specs

OSPool is highly heterogeneous because each contributing site donates whatever spare cycles it has at any given moment. CPU jobs are abundant and land on a wide range of Intel and AMD nodes. GPU availability is a strict subset of sites and a strict subset of nodes within those sites. The table below lists the GPU types most frequently observed on the OSPool dashboard during 2024–2026, with dense FP16 tensor throughput stated without sparsity for direct comparison against the 989 TFLOPS H200 SXM baseline.

GPU	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	Per-job limit	Preemption risk
A100 80GB	80	312	2.04	1 GPU	Low–medium
A100 40GB	40	312	1.55	1 GPU	Low–medium
A40	48	150	0.70	1 GPU	Medium
V100	16/32	125	0.90	1–2 GPU	Medium

GPU	VRAM (GB)	FP16 TFLOPS	BW (TB/s)	Per-job limit	Preemption risk
P100	16	21	0.73	1–2 GPU	Low
RTX 3090	24	142	0.94	1 GPU	High
RTX 2080 Ti	11	108	0.62	1–2 GPU	High
T4	16	65	0.32	1–2 GPU	Medium

The dominant GPU classes in steady state are V100, RTX 2080 Ti, and T4. A100 capacity exists but is contested, and H100 / H200 nodes are essentially absent from the opportunistic pool. Memory per CPU core typically caps at 8–16 GB, and per-job disk in `$_CONDOR_SCRATCH_DIR` is usually bounded near 20–75 GB depending on site.

Allocation Model

There is no fixed allocation. A user with an Access Point account submits HTCondor jobs and they begin matching against idle slots within minutes. Per-job constraints are imposed in lieu of a project quota. Wallclock is typically capped at 24 hours, with most sites preferring jobs under 10 hours so that preemption losses stay bounded. Memory is typically capped near 16 GB per core, jobs are single-node only (no MPI across nodes), and persistent storage is *not* provided — input and output staging is handled through the *Open Science Data Federation* (OSDF), which caches data near the execution site for the lifetime of the job.

GPU jobs are subject to additional limits. A user is normally restricted to one GPU per job and a small number of concurrent GPU slots (often 20–100 in practice for a single submitter, occasionally more). CPU throughput is far more elastic. A well-formed CPU workload can routinely saturate several thousand cores during a sustained burn, yielding tens of thousands of CPU-core-hours per day with no formal request. GPU-hours arrive more slowly and at a lower per-GPU tier than NAIRR, ACCESS, or commercial cloud.

Throughput Model and H200-hour Equivalence

Because the GPU mix is heterogeneous and concurrency depends on moment-to-moment supply, useful equivalence figures are *scenario-based* rather than fixed allocations. A reference upper bound is straightforward. One thousand concurrent A100 job-slots running for one wallclock hour is $1000 \times \frac{312}{989} \approx 315$ H200-hours per wallclock hour. That ceiling is rare. A more realistic steady-state for a single user is 20–100 concurrent GPU slots drawn mostly from V100 and RTX-class cards, with 10–30% of jobs preempted and re-run.

The table below sketches three weekly scenarios, assuming a representative GPU mix of 40% V100 (125 TFLOPS), 30% RTX (110 TFLOPS), 20% A100 (312 TFLOPS), and 10% T4 (65 TFLOPS), giving a blended 144 TFLOPS per slot, and netting out a 20% preemption tax.

Scenario	GPU job-hours/ wk	Blended TFLOPS	Preempt. tax	H200-hr eq./wk
Small individual user	1,000	144	20%	117
Medium sustained burn	10,000	144	20%	1,165
Large coordinated campaign	100,000	144	20%	11,650

CPU equivalence is not stated in H200-hours because CPU and tensor-GPU throughput are not interchangeable for ML workloads. For preprocessing, feature extraction, and Monte Carlo, the CPU pool can deliver the equivalent of several large clusters worth of cycles per week at no cost.

Approval Timeline

Onboarding takes approximately one week. The user fills a short registration form with an Access Point, declares a project name and brief description, identifies a faculty or staff sponsor, and receives a login on the AP. There is no per-project review, no allocation committee, and no ranking against other applicants. A first job typically runs the same day the account is activated.

Eligibility

Access is open to United States-based open-science researchers, including graduate students with advisor sign-off. Non-US collaborators may participate as members of a US-based principal investigator's project. Classified, ITAR-controlled, export-controlled, and HIPAA-restricted work is forbidden on OSPool, since the opportunistic execution sites cannot guarantee the data-handling controls those regimes require.

Things to Know

Several practical points dominate day-to-day use. Workloads must be refactorable into independent short jobs. There is no MPI across nodes, and jobs that exceed wallclock or memory caps are killed. Checkpointing is the user's responsibility, and any job longer than a few hours should write periodic state to OSDF so that a preemption loss costs minutes rather than hours. Singularity / Apptainer containers are strongly recommended, both because they make the job portable across heterogeneous sites and because they sidestep host library mismatches that would otherwise produce cryptic CUDA failures.

OSDF handles input and output staging but is not persistent storage. Datasets must be pushed to OSDF (or an HTTP/S3 origin federated into OSDF) before the campaign and pulled back to durable storage afterward. Preemption rates of 10–30% are normal and should be modeled into job sizing. The sweet spot is hyperparameter sweeps, Monte Carlo, ensemble inference, feature extraction over large datasets, and

genome-analysis-style pipelines. The pool is not appropriate for end-to-end training of a 7B-class language model, which would require multi-node NVLink-class fabric that the opportunistic pool does not expose.

Strategic Fit

OSPool is excellent *complementary* compute for any project that can be embarrassingly parallelized. It is poorly suited to tightly coupled multi-GPU foundation-model training. The recommended pattern is to train a model on NAIRR or ACCESS, then run sweeps, ablations, ensemble inference, and large-scale feature extraction on OSPool. CPU-side preprocessing of multi-terabyte datasets is also a natural fit, since the CPU pool is effectively unmetered for a well-formed submitter.

Summary

Workload type	H200-hr eq. /wk	Effort to port	Fit
Hyperparameter sweep (small models)	1,000–10,000	Low	Excellent
Ensemble / Monte Carlo inference	1,000–10,000	Low	Excellent
Feature extraction over large datasets	CPU-bound, very high	Low–medium	Excellent
Single-GPU fine-tuning ($\leq 7B$)	100–1,000	Medium (checkpoint)	Good
Multi-GPU single-node training	50–500	Medium	Marginal
Multi-node MPI / 7B+ pretraining	N/A	Not supported	Unsuitable

H200-hour equivalence is computed at dense FP16/BF16 tensor throughput (989 TFLOPS, 141 GB HBM3e, 4.8 TB/s) per H200 SXM GPU. Conversions below use the per-hour FP16 TFLOP ratio; memory-bound workloads scale differently and the bandwidth column is provided so readers can re-derive their own equivalence.

Sources

- OSG Consortium home and OSPool: osg-htc.org · ospool.osg-htc.org
- OSG Connect Access Point and onboarding: portal.osg-htc.org · OSG documentation
- OSPool dashboard and resource accounting: GRACC accounting · OSPool live dashboard
- OSDF data federation: OSDF service page · OSG data documentation
- HTCondor user manual (job submission, preemption, GPU requests): htcondor.readthedocs.io
- CHTC and contributing-site documentation: CHTC · Nebraska HCC · UCSD SDSC
- GPU specifications cross-checked against NVIDIA A100, A40, V100, P100, T4, RTX 3090, and RTX 2080 Ti datasheets.